# ECDLP transfer methodology: known-case validation

aburan28/crypto research record

2026-10-13

## Outcome

The fail-closed transfer methodology and native validator classify six known public cases consistently with their frozen source evidence. All six assessments pass the machine contract, and four deliberately invalid semantic controls pass by being rejected. Two fixtures support a valid subgroup transfer, two remain unknown at the transfer gate, and two are refuted. None supports a modeled or measured computational advantage.

The replay also found and repaired a pre-existing build defect in `tools/ecdlp-surfaces`: the checked-in binary module was not exported from the crate root, so the package's own validation script could not compile the n=83 checker. After adding the missing module export, the existing package reported 22 tests passed, zero failed, Clippy clean, and byte-for-byte replay of its committed examples.

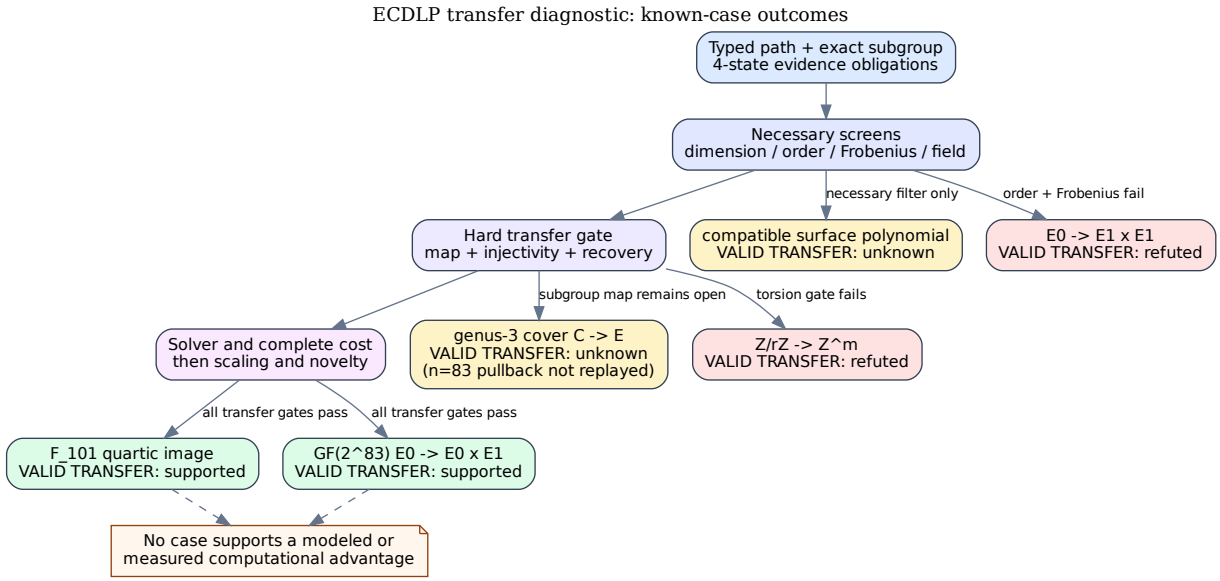



Figure 1: The diagnostic accepts complete transfers, preserves unknowns after merely necessary filters or unexecuted subgroup maps, and rejects algebraically impossible destinations.

## Requested scope and evidence

| Requirement | Status | Evidence |
|---|---|---|
| Put the methodology in `aburan28/crypto` | Verified complete | `docs/transfer-assessment/METHODOLOGY.md` |

| Requirement | Status | Evidence |
|---|---|---|
| Make it machine-checkable | Verified complete | `tools/transfer-assessment`, schema, four invalid controls |
| Run it on known transfers | Verified complete for six bounded public cases | `assessments/` and `validation-summary.json` |
| Include covers, surfaces, higher genus, and lattices | Verified complete as diagnostic coverage | quartic image, product surfaces, genus-3 cover, torsion-free lattice control |
| Establish a new faster ECDLP reduction | Not established and not implied | all advantage verdicts are `unknown` or `not_applicable` |
| Validate arbitrary covers/lattices/surfaces | Open beyond the six cases | each new candidate still needs its own identities, map, subgroup, solver, and cost evidence |

The validation profile is an assessment consistency check. It does not make an official state transition, independently prove every cited artifact, or turn template completion into evidence.

## Pinned basis

Repository revision: `3d4a891bd07000cf63935f5c876a22ff5e193315`.

The cases cite these exact paths at that revision:

- `tools/ecdlp-surfaces/examples/transfer_certificate.json`;
- `tools/ecdlp-surfaces/examples/frobenius_result.json`;
- `tools/ecdlp-surfaces/examples/binary_n83_a0.json`;
- `tools/ecdlp-surfaces/README.md`; and
- `research/hyperelliptic_cover_infrastructure_20261006/README.md`.

The lattice case is a known-false type control from the methodology: an ordinary additive lattice `Z^m` is torsion-free, so a homomorphism from the finite cyclic group `Z/rZ` has zero image. It does not reject finite lattice quotients, tori, endomorphism-kernel lattices, or relation lattices; those must be typed and audited separately.

## Exact classifications

| Case | Strongest established stage | Valid transfer | Algorithmically relevant | Modeled / measured advantage |
|---|---|---|---|---|
| Small-prime quartic image | explicit coordinate isomorphism, subgroup and recovery checks | supported | supported for generic group algorithms | unknown / unknown |
| n=83 inclusion `E_0 -> E_0 x E_1` | explicit factor inclusion and projection recovery | supported | supported for generic group algorithms | unknown / unknown |
| Compatible degree-four Frobenius polynomial | necessary isogeny-class filter only | unknown | unknown | unknown / unknown |

| Case | Strongest established stage | Valid transfer | Algorithmically relevant | Modeled / measured advantage |
|---|---|---|---|---|
| n=83 genus-3 cover | curve construction and `C -> E` map; exact n=83 Jacobian pullback unexecuted | unknown | unknown | unknown / unknown |
| n=83 proposed `E_0 -> E_1 x E_1` | order and Frobenius obstruction | refuted | refuted | not applicable |
| proposed `Z/rZ -> Z^m` | torsion obstruction | refuted | refuted | not applicable |

This is the intended discrimination:

1. A faithful representation may be a valid transfer without an advantage.
2. Frobenius compatibility is necessary evidence, not a homomorphism.
3. A curve map `C -> E` is not the required executable subgroup map `E -> J(C)`; the n=83 pullback remains unknown until replayed.
4. Group-order, Frobenius, and torsion contradictions reject impossible paths before solver work.

## Native validator and controls

`tools/transfer-assessment` parses every assessment, binds each supported or refuted state to a declared source id, and enforces cumulative verdict gates:

- `valid_transfer=supported` requires supported map correctness, subgroup preservation, and scalar recovery;
- `algorithmically_relevant=supported` also requires a named applicable solver;
- modeled advantage additionally requires complete accounting;
- measured advantage additionally requires scaling support and measured data;
- novelty cannot be promoted without the earlier cumulative verdicts; and
- the diagnostic profile refuses an official status change.

Four Rust tests exercised one valid unknown record, one fully supported transfer, promotion without the three transfer gates, and unbound evidence plus an unsupported measured-advantage promotion. Result: 4 passed, 0 failed.

The six study assessments then produced `transfer-assessment-validation-summary-v1` with `assessments_checked=6`, `all_valid=true`, and no validation errors.

## Reproduction

Environment:

```
Linux 6.18.44 x86_64 GNU/Linux
rustc 1.99.0 (b940084d7 2026-09-28)
cargo 1.99.0 (5f94df478 2026-08-27)
Graphviz 2.42.4
Pandoc 3.1.11.1
```

Commands:

```
cargo test --release --locked \
  --manifest-path tools/transfer-assessment/Cargo.toml

cargo run --release --locked \
```

```
  --manifest-path tools/transfer-assessment/Cargo.toml -- \
  --output research/transfer-methodology-validation-20261013/validation-summary.json \
  research/transfer-methodology-validation-20261013/assessments/*.json

tools/ecdlp-surfaces/check.sh

cargo test --release --lib --quiet
```

The first surface-package replay at the pinned base failed during compilation:

```
error[E0432]: unresolved import `ecdlp_surfaces::binary`
```

After `pub mod binary;` was restored in `tools/ecdlp-surfaces/src/lib.rs`, the same command passed 9 binary tests and 13 math tests, Clippy, and committed example regeneration. The repair changes module reachability only; the fixture mathematics and frozen JSON were not edited.

The final repository-wide run, with all compile-time fixtures materialized and the repository's local-socket tests permitted, reported 4,058 passed, zero failed, and 114 ignored in 217.16 seconds. The first sparse/sandboxed attempt reported 4,043 passed and 15 failed: six failures were missing sparse-checkout fixtures, and nine were local-socket `Operation not permitted` errors. After materializing the exact two remaining fixture directories, the six affected tests passed; all 37 `pollard_collab` tests passed with local sockets enabled; then the clean full run above passed. These were access/setup failures, not mathematical evidence and not changes to the tested implementation.

## Limits and next discriminating check

This round validates methodology behavior on bounded known cases, not general completeness. It contains no performance benchmark, so no `benchmarks/` entry or PNG performance chart is required. No new or substantially revised mathematical theorem is asserted, so there is no new Lean theorem to map.

The next check for the higher-genus route is concrete: execute the native n=83 `OrdinaryBinaryCover` pullbacks for the exact pinned `P` and `Q`, verify image order `r`, the degree-three pull/push composition including exceptional fibers, and source scalar recovery. A passing transfer still would not establish an advantage until a destination solver and complete matched cost comparison are attached.

Conductor scope coordination could not be performed: `conductor` was absent from this execution environment and the required command exited 1 with no output. Work was kept to `docs/transfer-assessment/`, `tools/transfer-assessment/`, this study directory, and the one-line surface module repair disclosed above.

## Graph scope

The boundary ledger, index-calculus scoreboard, progress timeline, curve registry, cover catalog, and existing surface diagrams were checked for applicability. None changes: this PR adds no new curve identity, map edge, admitted performance measurement, solver advantage, or official claim. The study-local diagram records only diagnostic classifications.